MIL-I-22834A(SHIPS)
12 April 1962
SUPERSEDING
MIL-I-22834(SHIPS)
14 March 1961

MILITARY SPECIFICATION

INSULATION, ELECTRICAL, DIELECTRIC BARRIER, LAMINATED, PLASTIC FILM AND SYNTHETIC FIBER MAT

1. SCOPE

1.1 Scope.- This specification covers insulation used primarily as dielectric barriers in class B (130°C) electrical machinery.

1.2 Classification

1.2.1 Compositions.- The electrical insulation shall be of the following compositions, as specified (see 6.3):

Composition D100 - One layer of polyethylene terephthalate fiber mat laminated to each side of a polyethylene terephthalate film or sheet (triplex). The mat is completely impregnated with epoxy resin.

Composition D70 - One layer of polyethylene terephthalate fiber mat laminated to each side of a polyethylene terephthalate film or sheet (triplex). The mat is 70 percent impregnated with epoxy resin.

Composition D - One layer of polyethylene terephthalate fiber mat laminated to each side of a polyethylene terephthalate film or sheet (triplex). The mat is heat bonded to the film.

1.2.2 Types.- The electrical insulation shall be of the following types (see 6.3):

Types:
- D100-323
- D100-353
- D100-535
- D70 -323
- D70 -353
- D -313
- D -323
- D -555
- D -636

1.2.3 Forms.- The electrical insulation shall be of the following forms, as specified (see 6.3):

Form R - Rolls, trimmed, 36 inches in width, in lengths as follows:

Type	Length per roll
	(Lineal yards)
D100-323	400
D100-353	300
D100-535	250
D70 -323	400
D70 -353	300
D -313	400
D -323	300
D -555	250
D -636	250

Form S - Sheets, 36 inches wide by 36 inches in length, in packages of 200 sheets.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

VV-I-530 - Insulating Oil, Electrical, for Transformers, Switches, and Circuit Breakers.

MILITARY

MIL-I-631 - Insulation, Electrical, Synthetic-Resin composition, Non-rigid.

FSC 5970

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STANDARDS

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York)

3. REQUIREMENTS

3.1 Material.-

3.1.1 Plastic film or sheet.- The polyethylene terephthalate film or sheet shall conform to type G of MIL-I-631.

3.1.2 Synthetic fiber mat.- The mat shall be composed of polyethylene terephthalate fibers and may be impregnated with epoxy resin. If the impregnation is complete with all fibers securely anchored, the mat shall be considered 100 percent impregnated, and shall be denoted D100. If the impregnation is only 70 percent complete, there will be freely moving fibers on the surface with a fuzzy appearance, and the mat will be denoted D70. If the mat is bonded to the film by heat bonding, and with freely moving fibers on the surface with a fuzzy appearance and with no impregnation, the mat will be denoted D.

3.1.3 Composite insulation.- The composite insulation shall be uniform in density without breaks, cracks, wrinkles, or flaws of any kind. The mat shall be uniform, and there shall be no tendency towards splitting or separation of the laminations. The layers shall be cemented together with either a class B flexible insulating binder, or by heat bonding. The insulation shall be free from electrically conductive particles, dirt, and other foreign material that may affect the serviceability of the finished product.

3.2 Construction and thickness.- The electrical insulation shall conform to the construction and thickness shown in table I (see 6.3).

Table I - Construction and thickness.

Type	Nominal combined insulation thickness (see 6.3)	Tolerance of average combined thickness from nominal thickness		Nominal thickness (inch) and component (designation) 1/		
		Minus	Plus	First layer	Second layer	Third layer
	Inch	Inch	Inch			
D100-323	0.009	0.001	0.001	0.003D100	0.002F	0.003D100
D100-353	.012	.001	.001	.003D100	.005F	.003D100
D100-535	.014	.001	.001	.005D100	.003F	.005D100
D70 -323	.009	.001	.001	.003D70	.002F	.003D70
D70 -353	.012	.001	.001	.003D70	.005F	.003D70
D -313	.007	.001	.001	.003D	.001F	.003D
D -353	.011	.001	.001	.003D	.005F	.003D
D -555	.015	.0015	.0015	.005D	.005F	.005D
D -636	.015	.0015	.0015	.006D	.003F	.006D

1/ Designations:
F - Plastic film or sheet.
D100 - Polyethylene terephthalate fiber mat completely impregnated with epoxy resin.
D70 - Polyethylene terephthalate fiber mat 70 percent impregnated with epoxy resin.
D - Polyethylene terephthalate fiber mat with no impregnation.

3.3 Physical requirements.- The electrical insulation shall conform to the physical requirements shown in table II.

3.4 Workmanship.- Workmanship shall be first class in every respect.

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Table II—Physical properties.

Property	Condition		Unit of value	Value required								
				Type D100-323	Type D100-353	Type D100-535	Type D70-323	Type D70-353	Type D-636	Type D-313	Type D-353	Type D-555
Dielectric strength	C-96/23/50	Straight	Volts per mil (minimum average)	650	1000	550	570	1000	600	425	1200	1050
	C-96/23/96	Straight	Volts per mil (minimum average)	625	975	525	550	975	600	425	1200	1050
	C-96/23/50	Bent lengthwise	Volts per mil (minimum average)	650	1000	550	570	1000	525	350	1100	1000
	C-96/23/50	Bent crosswise	Volts per mil (minimum average)	650	1000	550	570	1000	525	350	1100	1000
	E-168/150	Bent lengthwise	Volts per mil (minimum average)	650	1000	550	570	1000	525	375	1100	1000
Volume resistivity	C-96/23/96	----------	Ohm-cms. (minimum average)	10^{13}	10^{13}	10^{13}	10^{13}	10^{13}	10^{13}	10^{13}	10^{13}	10^{13}
Tensile strength	C-96/23/50	Lengthwise	Pounds per inch width (minimum average)	115	140	185	50	115	65	20	110	115
	C-96/23/50	Crosswise	Pounds per inch width (minimum average)	60	105	95	30	100	60	20	110	115
Tear resistance (slot cell)	C-96/23/50	Lengthwise	Pounds (minimum average)	20	45	35	12	32	17	5	33	35
	C-96/23/50	Crosswise	Pounds (minimum average)	15	40	30	9	30	15	5	33	35
Delamination	C-96/180	----------	----------	None	None	None	None	None	None	None	None	None
Absorbency	C-96/23/50	----------	seconds (maximum average)	----	----	----	----	----	80	200	200	100

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4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.-

4.2.1 Lot.- All material of the same composition, type, and form, manufactured under essentially the same conditions within a period of 15 days, and offered for delivery at the same time, shall be considered a lot for purposes of quality conformance inspection.

4.2.2 Sampling for visual and dimensional examination.- Samples of material shall be selected from each lot for examination of visual and dimensional characteristics in accordance with table III. The lot size and sample size shall be stated in terms of rolls and packages. Approximately 5 yards shall be examined from any roll or package.

Table III - Sampling for visual and dimensional examination.

Lot size number of rolls and packages	Sample size rolls and packages to be examined	Acceptance number (defective)	Rejection number (defective)
2 to 8	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 40	7	0	1
41 to 65	10	0	1
66 to 110	15	0	1
111 to 180	20	1	2
181 to 300	25	1	2
301 to 500	30	1	2

.2.3 Sampling for quality conformance tests.- One fifth of the rolls or packages selected for the visual and dimensional examination shall be subjected to the quality conformance tests specified in table IV; however, the smallest number of rolls or packages selected for any sample shall be two.

4.3 Quality conformance inspection.-

4.3.1 Visual and dimensional examination.- Each of the samples selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify conformance with the requirements of 3.1, 3.2, and 3.4. Any sample roll or package containing one or more visual or dimensional defects shall be considered defective, and if the number of defective rolls or packages exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample. In the case of rolls, the outer layer of the rolls shall be discarded, and the next five linear yards shall be examined.

4.3.2 Quality conformance tests.- Quality conformance tests shall be conducted on samples of material selected in accordance with 4.2.3. Two linear yards from each sample roll or package shall be subjected to the tests specified in table IV. Results shall be averaged for each sample roll or package represented to determine conformance with this specification. If the average for any sample roll or package fails to conform with any requirement, this shall be cause for rejection of the lot represented by the sample.

4.4 Test conditions.-

4.4.1 Conditions.- Unless otherwise specified herein, all tests shall be conducted under conditions of 23° ± 1.1° centigrade (C.) and 50 ± 2 percent relative humidity (r.h.).

4.4.2 Designation.- Conditioning procedures shall be designated as follows:

(a) A capital letter indicating the general condition of the specimen: "C" indicates humidity and "E" indicates temperature conditioning.

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(b) A number indicating in hours the duration of the conditioning.
(c) A number indicating in degrees C. the conditioning temperature.
(d) A number indicating relative humidity whenever relative humidity is controlled.

The numbers shall be separated from each other by slant marks, and from the capital letter by a dash.

Examples:
C-96/23/50 - 96 hours, 23°C., 50 percent relative humidity.
C-96/23/96 - 96 hours, 23°C., 96 percent relative humidity.
E-168/150 - 168 hours, 150°C.

4.5 Test procedures. -

4.5.1 Thickness. - The total thickness of the material shall be determined by the use of a dial type micrometer. The lower or fixed anvil of the dial micrometer shall consist of a flat steel plate 2 inches in diameter; the upper or movable member shall consist of a foot having a flat steel face measuring 1/4 inch in diameter, and shall be arranged so as to move with its face exactly parallel to the face of the anvil at all times. The movable member shall be of such weight, and be attached to the operating rod of the micrometer in such a manner, that 20 ounces dead weight shall be applied to all thicknesses of samples being measured. The thickness shall be determined as the average of at least 10 readings taken at representative points on the material. The maximum, minimum, and average values shall be reported.

4.5.2 Dielectric strength. -

4.5.2.1 Test equipment. - The test equipment shall consist of a high voltage transformer rated at not less than 2 kilovolt-amperes and the necessary auxiliary equipment for applying, controlling, and measuring the test voltage. The power supply shall consist of an alternating-current source having as nearly a true sine wave as possible at a frequency not exceeding 100 cycles per second.

4.5.2.2 Voltage control. - The high-tension voltage taken from the secondary of the testing transformer shall be capable of being raised gradually from any point, and in no case more than 500 volts at a step. The control may be made by generator field regulation with an induction regulator, or with a variable ratio autotransformer. Any method of controlling the voltage which does not distort the wave more than 10 percent from a sinusoidal shape is satisfactory.

4.5.2.3 Voltage measurement. - The voltage may be measured by any method which gives root-mean-square values, preferably by means of a voltmeter connected to a tertiary coil in the testing transformer, or to a separate step-down instrument potential transformer. A voltmeter on the low tension side of the transformer is satisfactory if the ratio of the transformation does not change under any test condition. An electrostatic voltmeter properly calibrated in the high tension circuit is also satisfactory. A spark gap may be used to check the readings at very high potentials.

4.5.2.4 Medium and rate of voltage rise. - The tests shall be conducted in air by the short-time method. Starting at zero, the voltage shall be increased uniformly to a breakdown, at a rate of 500 volts per second.

4.5.2.5 Procedure - straight specimens. - A sufficient number of specimens 4 inches by 4 inches shall be cut to permit making 10 breakdowns after each of the following conditions:

C-96/23/50 (on straight specimens).
C-96/23/96 (on straight specimens).

The test electrodes shall consist of metal cylinders 2 inches in diameter and not less than 1/2 inch in length with edges rounded to a radius of 1/4 inch. The total weight on the sample shall be approximately 1 pound.

4.5.2.6 Procedure - bent specimens. - A sufficient number of specimens 3 by 3-1/2 inches shall be cut to permit making 10 breakdowns after each of the following conditions:

C-96/23/50 (on specimens bent lengthwise).
C-96/23/50 (on specimens bent crosswise).
E-168/150 (on specimens bent lengthwise).

The specimens shall be bent 90 degrees in the V-shaped jig shown on figure 1. Breakdowns shall be made using 90-degree, V-shaped electrodes designed to hold the specimens in a bent position. These electrodes are shown on figure 2. All edges of the bending jig and electrodes are rounded to approximately 1/64 inch radius. Oven aged specimens shall be clamped on a rectangular aluminum channel to hold them in a bent position during aging. After aging, the channels shall be removed from the oven and allowed to return to room condition. The specimens shall then be removed from the channel and short-time breakdowns made using the V-shaped electrodes.

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Table IV - Quality conformance tests.

Test	Requirement paragraph	Test reference	Minimum number of determinations per sample
Thickness, condition C-96/23/50	3.2	4.5.1	10
Dielectric strength: Straight, condition C-96/23/50	3.3	4.5.2	10
Straight, condition C-96/23/96	3.3	4.5.2	10
Bent lengthwise, condition C-96/23/50	3.3	4.5.2	10
Bent crosswise, condition C-96/23/50	3.3	4.5.2	10
Bent lengthwise, condition E-168/150	3.3	4.5.2	10
Volume resistivity, condition C-96/23/96	3.3	4.5.3	3
Tensile strength: Lengthwise, condition C-96/23/50	3.3	4.5.4	5
Crosswise condition C-96/23/50	3.3	4.5.4	5
Tear resistance (slot cell): Lengthwise, condition C-96/23/50	3.3	4.5.5	5
Crosswise, condition C-96/23/50	3.3	4.5.5	5
Delamination, condition E96/180	3.3	4.5.6	3
Absorbency, condition A	3.3	4.5.7	5

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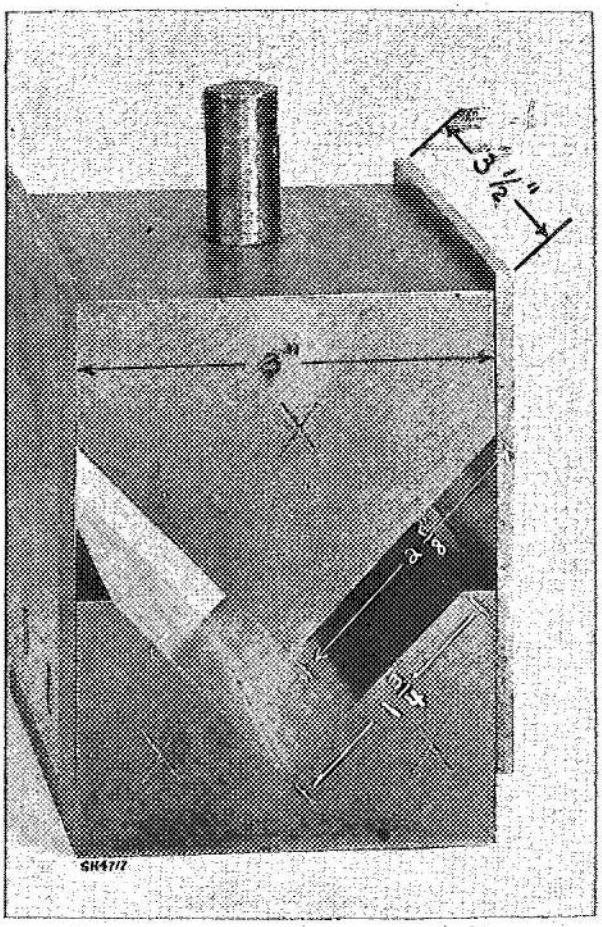


Figure 1 - Bending jig 90° v-shaped

4.5.2.7 Calculation of dielectric strength. - The dielectric strength shall be computed by dividing the average dielectric breakdown by the average thickness, and shall be reported in volts per mil. The average thickness shall be determined as specified in 4.5.1.

4.5.3 Volume resistivity. -

4.5.3.1 Apparatus. - The apparatus shall consist of a source of direct-current potential of 100 to 200 volts, a galvanometer with suitable shunts having a minimum sensitivity of 0.0005 microampere per millimeter at 1 meter-scale distance, a calibrated resistance, reversing switches, and keys. Any other apparatus may be used provided the same accuracy is obtained and the same voltage is applied.

4.5.3.2 Test specimens. - The size of the test specimens shall be 4 by 4 inches. Three specimens shall be tested after Condition C-96/23/96.

4.5.3.3 Test electrodes. - One electrode shall be of metal foil, 3 inches in diameter, on one surface of the specimen. The electrode on the other surface shall be of metal foil 2-5/8 inches in diameter. A concentric, metal-foil guard ring electrode shall be used, and shall be 1/8 inch wide and 3 inches in outside diameter, and be spaced 1/16 inch from the inner electrode. The electrodes shall be applied to the specimen by means of a very thin film of petrolatum, and shall be thoroughly rolled to exclude air. The electrodes shall be accurately centered with respect to each other.

4.5.3.4 Procedure. - The volume resistance between the guarded and unguarded electrodes shall be measured after voltage has been applied for 1 minute.

4.5.3.5 Calculation. - Volume resistivity, r, in ohm-centimeters, shall be calculated as follows:

$$r = \frac{RA \times 10^6}{t}$$

Where:
R = volume resistance in megohms.
A = area of guarded electrode in square centimeters.
t = average thickness of specimen in centimeters.

The average volume resistivity shall be reported.

4.5.4 Tensile strength. -

4.5.4.1 Apparatus. - Tensile strength shall be determined on an approved pendulum-type or constant rate of jaw travel testing machine. The machine shall be power-driven.

4.5.4.2 Specimens. - The test specimens shall be cut using die I of method 4111 of FED-STD-601. The surface of the specimen shall be smooth and free from scratches. Five specimens cut lengthwise and five cut crosswise of the sheet shall be used after condition C-96/23/50.

4.5.4.3 Procedure. - The test specimens shall be placed in the testing machine and stretched until they break. The rate of separation of the jaws of the machine shall be 12 plus or minus 1/2 inches per minute. The tensile strength shall be reported in pounds per inch width.

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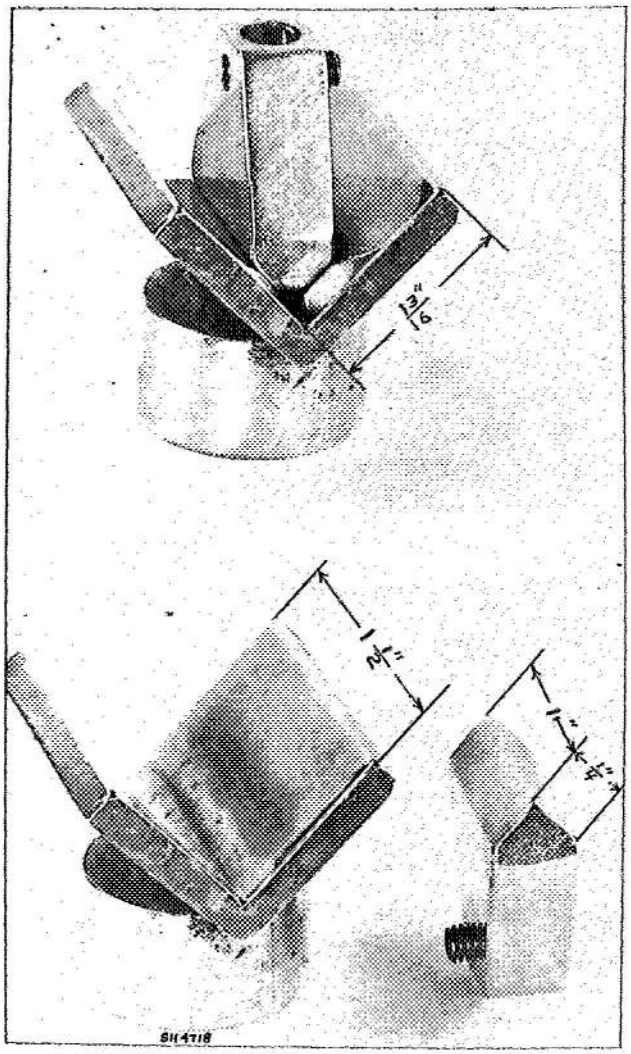


Figure 2 - Dielectric test electrodes 90° v-shaped

4.5.5 Tear resistance (slot-cell). -

4.5.5.1 Apparatus. - The slot-cell tear resistance shall be determined using the tensile strength apparatus specified in 4.5.4.1 with specimens mounted in the slot-cell tear resistance jig shown on figure 3.

NOTE: Working drawings of this jig are available from Material Laboratory (Code 955), New York Naval Shipyard, Brooklyn 1, N. Y.

4.5.5.2 Specimens. - Five specimens shall be torn in the tearing jig in the lengthwise direction and five specimens shall be torn in the tearing jig in the crosswise direction after condition C-96/23/50. The directions shall be reported as the directions in which the tears run. The specimens shall be 2-1/2 by 3 inches. The specimens shall be inserted and aligned in the tearing jig. The specimen holder shall be clamped to the upper jaw of the tensile machine. The tearing plate together with the attached guide plate shall then be clamped to the lower jaw. The tearing edges which come in contact with the specimen have a radius of curvature of 1/32 inch.

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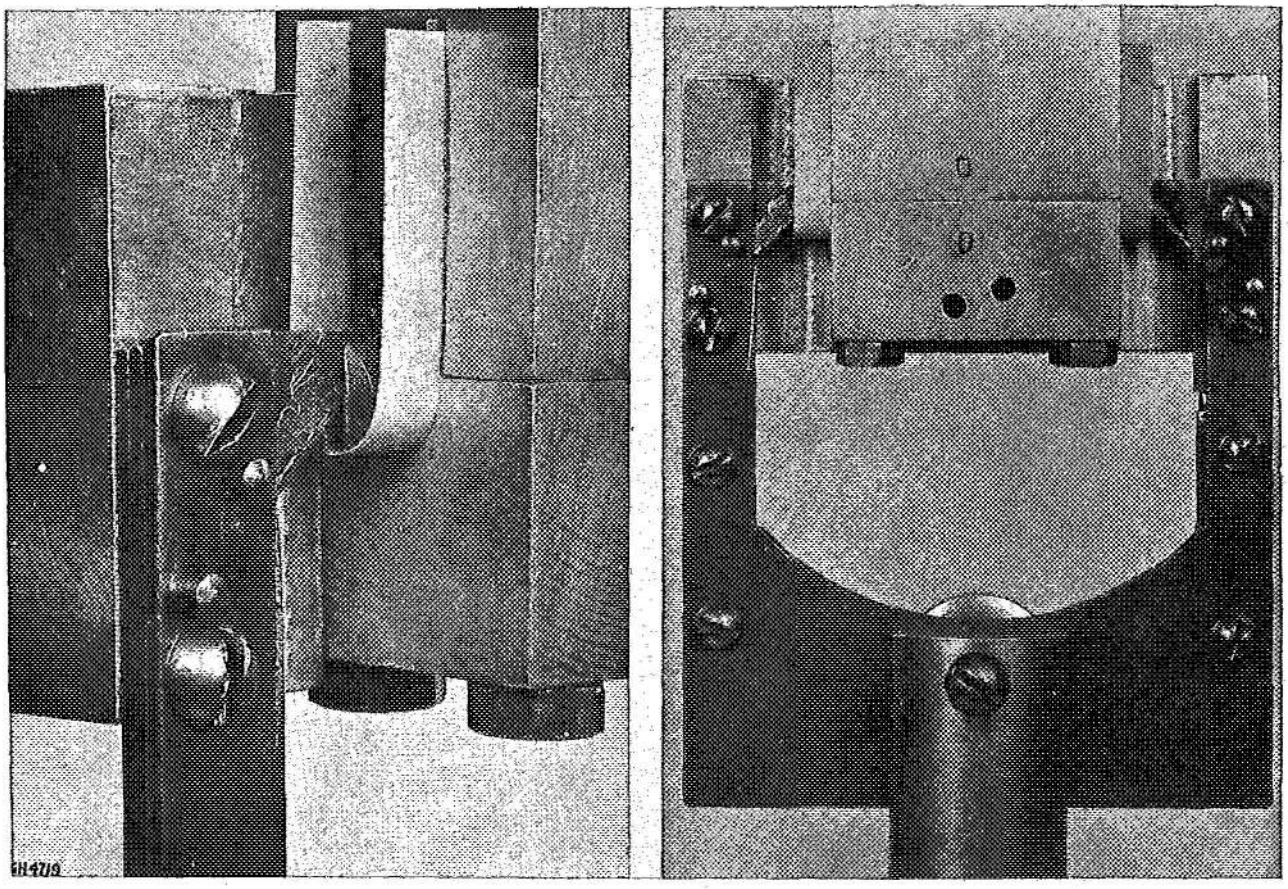


Figure 3 - Slot-cell tear resistance jig specimen in position to start tear

5.2.1.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The materials covered by this specification are class B electrical insulations with a continuous operating temperature limit of 130°C. They are intended for use as slot-cell insulations and other dielectric barriers in electrical machinery where excellent resistance to high humidity is required. Excellent bonding with epoxy resins may be achieved in encapsulated equipment when this material is used.

6.2 Other constructions. - Other type constructions and thicknesses than those specified (see 1.2.2) may be utilized by equipment suppliers; however, the material supplier should provide substantiating test data indicating that the material was tested using the test methods described herein. The test data values should be acceptable to the equipment supplier, and approved by the bureau or agency concerned.

6.3 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Composition required (see 1.2.1).
(c) Type required (see 1.2.2).
(d) Form required (see 1.2.3).
(e) Length of roll in yards, if to be furnished in form R (see 1.2.3).
(f) Nominal thickness of combined insulation and each component (see 3.2).
(g) Preparation for delivery if other than as specified in 5.1 (see 5.2).

6.4 Approximate areas per pound. - The approximate area per pound of the materials covered by this specification are shown in table V.

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4.5.5.3 Procedure. - The maximum force in pounds developed in tearing through the edges of the specimen extending beyond the holder at a constant rate of 12 inches per minute shall be recorded on a chart. The average tear resistance in pounds for each direction of tear shall be calculated and reported.

4.5.6 Delamination. - Three specimens shall be cut 4 by 4 inches and conditioned in an oven condition E-96/180, removed from the oven, and cooled to 23° C. at 50 percent relative humidity. The specimens shall be visually examined for delamination.

4.5.7 Absorbency. - Five specimens shall be cut 4 by 4 inches. Each specimen shall be tested by placing the sample on a horizontal surface with an overhead light source available. A 1-milliliter measuring pipet graduated in hundredths of a milliliter shall be filled with transformer oil conforming to VV-I-530. The test shall be made at 23° + 2° C. With the tip of the pipet 1 inch above the specimen, allow 0.02 ml (one drop) of oil to flow on the specimen near its center. With a stop watch, measure the rate of absorption in seconds from the start of the flow of oil until the drop of oil is completely absorbed. This is indicated by the termination of reflection of incident light from the surface. Five specimens shall be tested, and the average of the times shall be reported.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use. -

5.1.1 Packaging. - Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early use.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name, and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.3).

5.2.1 The following provides various levels of protection during domestic shipment and storage or over seas shipment, which may be required when procurement is made (see 6.3).

5.2.1.1 Packaging. -

5.2.1.1.1 Level A. - Rolls and sheets shall be packaged as follows:

5.2.1.1.1.1 Rolls. - When furnished in rolls, the material shall be wound tightly on chipboard or fiber tubes. The tube should be in one continuous length running the full width of the roll, and shall have a minimum wall thickness of 1/4-inch. The inside diameter of the tube shall be not less than 2-1/2 inches nor more than 3-1/2 inches. A tight-fitting wood plug having a 1/2-inch diameter hole shall be provided in each end of the tube. Material shall be rolled straight so that edges will form an even surface at each end of the roll. Each roll shall be wrapped with minimum 60-pound basis-weight (24 X 36-500 sheets) kraft paper. A minimum of 2-inch overlap should be provided at all overlapping edges. All seams, joints, and closures of the wrap shall be sealed with adhesive or gummed kraft tape.

5.2.1.1.1.2 Sheets. - Each package of 200 sheets (see 1.2.3) shall be overwrapped as specified for rolls (see 5.2.1.1.1.1.)

5.2.1.1.2 Level C. - Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity for early use.

5.2.1.2 Packing. -

5.2.1.2.1 Levels A and B. -

5.2.1.2.1.1 Rolls. - Rolls shall be individually packed in snug-fitting fiber drums conforming to PPP-D-715 as follows:

Type II drums - Level A packing.
Type I drums - Level B packing.

5.2.1.2.1.2 Sheets. - Sheets shall be packed flat in fiberboard box is conforming to PPP-B-636, with the exception of limitations on the weight of box and contents, and as follows:

Class 1 boxes - Level B packing.
Class 2 boxes - Level A packing.

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Table V - Areas per pound (approximate).

Type	Nominal combined insulation thickness	Area in square yards per pound
	Inch	
D100-323	0.009	2.40
D100-353	.012	1.62
D100-535	.014	1.40
D70 -323	.009	2.64
D70 -353	.012	1.72
D -636	.015	2.12
D -313	.007	3.10
D -353	.011	2.0
D -555	.015	1.6

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 5970-N039Sh)

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